

C-CAT SECTION B — NETWORKING FUNDAMENTALS

Topics 10 – 17

High-Intensity Revision Guide

Networks • Topologies • Devices • Media • Switching • Performance • Internet • Security

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TOPIC 10

Introduction to Computer Networks

LAN • WAN • MAN • PAN • Internet • Intranet • Extranet

What is a Computer Network?

A **computer network** is a collection of two or more computing devices connected together so they can **share data, resources (printers, storage, internet), and communicate** with each other.

- **Node** — any device on the network: computer, phone, printer, server.
- **Link** — the connection (wired or wireless) between two nodes.
- **Protocol** — agreed set of rules that devices follow to communicate correctly.
- **Bandwidth** — the maximum data-carrying capacity of a link (measured in bits per second).

Why Networks Exist — Problems Solved

- **Resource Sharing** — one printer shared by 50 computers instead of 50 printers.
- **Data Sharing** — files accessible from any machine on the network.
- **Communication** — email, chat, video calls between devices.
- **Centralized Management** — updates applied once on server; all devices benefit.
- **Cost Reduction** — share expensive hardware and internet connections.

Types of Networks by Geographic Area

Type	Full Name	Coverage Area	Example
PAN	Personal Area Network	~10 metres (personal space)	Bluetooth between phone and earbuds
LAN	Local Area Network	Single building / campus	Office network, school lab, home WiFi
MAN	Metropolitan Area Network	City / large campus	City-wide cable TV network, university across campuses
WAN	Wide Area Network	Countries / continents / global	The Internet, bank branch connections across India

SCALE DIAGRAM (small to large):

```
[Phone] <--Bluetooth--> [Earbuds] PAN (~10 m)
[PC1]--[Switch]--[PC2] LAN (1 building)
|
[PC3]
[Office A]----[MAN backbone]----[Office B] MAN (1 city)
[India HQ]====[Internet/Leased Line]====[US Branch] WAN (global)
```

Internet vs Intranet vs Extranet

Type	Definition	Who Can Access	Example
Internet	Global public network of millions of interconnected networks	Anyone worldwide	Browsing google.com

Intranet	Private network INSIDE an organization, using internet technologies (web pages, email) but restricted to internal users only	Only employees/members inside the org	Company internal HR portal
Extranet	Extension of an intranet that allows SPECIFIC outsiders (partners, vendors) controlled access	Internal staff + authorized external partners	Supplier portal where vendor checks orders

INTERNET vs INTRANET vs EXTRANET:

```

+-----+
| INTERNET (public) |
| +-----+ |
| | EXTRANET (controlled access) | | | |
| | +-----+ | |
| | | INTRANET (private) | | |
| | | [HR] [Finance] [Dev] | | |
| | +-----+ | |
| | + Authorized Partners | |
| +-----+ |
+-----+

Think: Intranet INSIDE Extranet INSIDE Internet

```

EXAM FOCUS

- PAN < LAN < MAN < WAN in terms of geographic coverage.
- Internet = public global network. Intranet = private internal network. Extranet = intranet extended to trusted outsiders.
- LAN is fastest (shortest distance, controlled media). WAN is slowest (long distances, many hops).
- Home WiFi = LAN. Bluetooth headphones = PAN. Internet = WAN.

COMMON EXAM TRAPS

- Intranet uses the SAME technologies as the Internet (HTTP, web browsers) but is PRIVATE and not accessible from outside.
- MAN is often confused with WAN — MAN covers a CITY; WAN covers countries/globe.

TOPIC 11

Network Topologies

Bus • Star • Ring • Mesh • Tree • Hybrid

What is Network Topology?

Network Topology is the arrangement / layout of nodes and links in a network — both physically (how cables are laid) and logically (how data flows). The choice of topology affects cost, performance, fault tolerance, and scalability.

1. Bus Topology

- All devices connect to a SINGLE central cable called the **backbone** or **bus**.
- Data travels in both directions along the bus. A **terminator** at each end absorbs signals to prevent reflection.
- **Advantage:** Easy to install, cheap, uses less cable.
- Data sent by one node is seen by ALL nodes — only the intended recipient accepts it.
- **Disadvantage:** If backbone cable breaks, ENTIRE network fails. Performance degrades with more devices. Difficult to troubleshoot.

BUS TOPOLOGY:

```
[T]---[A]---[B]---[C]---[D]---[E]---[T]
```

```
| |
```

Terminator Terminator

(absorbs signal) (absorbs signal)

T = Terminator | A,B,C,D,E = Nodes (computers)

All share ONE backbone cable.

If backbone breaks anywhere --> ALL nodes disconnected.

2. Star Topology

- All devices connect to a **central hub or switch**. No device connects directly to another.
- Data from one node goes TO the hub/switch first, which then forwards it to the destination.
- **Advantage:** If one cable or node fails, only THAT node is affected — rest of network works fine. Easy to add/remove devices. Easy to troubleshoot.
- **Disadvantage:** If the central hub/switch fails, the ENTIRE network goes down. Requires more cable than bus.
- Most common topology in modern homes and offices (your WiFi router is the center).

STAR TOPOLOGY:

```
[A]
```

```
|
```

```
[B]---[HUB/SWITCH]---[C]
```

```
|
```

```
[D]
```

```
|
```

```
[E]
```

Every node connects ONLY to the central hub/switch.

If [B]'s cable breaks --> only B is affected.

If [HUB] fails --> ALL nodes lose connection.

3. Ring Topology

- Each device connects to exactly **two other devices**, forming a closed loop (ring).
- Data travels in ONE direction (unidirectional) or both directions (bidirectional) around the ring.
- A special signal called a **token** circulates the ring — a device can only send data when it holds the token (**Token Ring** protocol).
- **Advantage:** Orderly data transfer, no collisions, performs well under heavy load.
- **Disadvantage:** If one node or cable fails, entire ring may break (unless dual-ring). Slower than star for small networks.

RING TOPOLOGY:

```
[A]
/ \
[E] [B]
| |
[D]----[C]
Data flows: A --> B --> C --> D --> E --> A (circular)
Token travels around the ring.
Only the node holding the TOKEN can transmit.
If C fails --> communication between B and D is broken.
```

4. Mesh Topology

- Every device connects to **every other device** directly (Full Mesh), or at least multiple devices (Partial Mesh).
- **Full Mesh formula:** Number of links = $n(n-1)/2$, where n = number of nodes.
- **Advantage:** Highest fault tolerance — if one link fails, data takes another route. Most reliable topology.
- **Disadvantage:** Very expensive — requires huge amount of cable and ports. Complex installation.
- Used where reliability is critical: military networks, backbone internet connections.

FULL MESH TOPOLOGY (4 nodes = 6 links):

```
[A]-----[B]
| \ / |
| \ / |
| /\ |
| / \ |
| / \ |
[D]-----[C]
Links: A-B, A-C, A-D, B-C, B-D, C-D =  $4 \times 3 / 2 = 6$  links
PARTIAL MESH (not every node connected to all):
[A]---[B]---[C]
\ /
[D]-----[E]
Some redundant paths but not full connection.
```

5. Tree Topology (Hierarchical)

- A **hierarchical arrangement** — nodes are arranged like branches of a tree with a **root node** at the top.
- Combination of bus and star topologies — star networks connected to a common bus backbone.
- Has a parent-child relationship: root → branch nodes → leaf nodes.

- **Advantage:** Easy to expand (add branches). Easy to manage by segments.
- **Disadvantage:** If root node fails, large sections of network fail. More cabling needed.
- Used in: large organizations with departments and sub-departments.

```

TREE TOPOLOGY:
[ROOT]
/ \
[Branch1] [Branch2]
/ \ / \
[Leaf1][Leaf2] [Leaf3][Leaf4]
Root = Main switch/router
Branch = Department-level switches
Leaf = Individual computers
Think: Company HQ --> Floor Switches --> Desktops

```

6. Hybrid Topology

- A **combination of two or more different topologies** — e.g., Star + Ring, or Star + Bus.
- Designed to take the advantages of each topology while minimizing their disadvantages.
- **Advantage:** Flexible, scalable, fault tolerant where needed.
- **Disadvantage:** Complex design and expensive to install.
- Most **real-world enterprise networks** are hybrid topologies.

```

HYBRID TOPOLOGY (Star + Bus example):
[BUS BACKBONE]-----
| |
[Hub/Switch] [Hub/Switch]
/ | \ / \
[A] [B] [C] [D] [E]
Two STAR networks connected via a BUS backbone.
Real networks combine topologies as needed.

```

Topology	Shape	Failure Impact	Cost	Use Case
Bus	Single backbone line	Whole network if backbone fails	Low	Old small offices
Star	All to central hub	Only 1 node if cable fails; all if hub fails	Medium	Homes, modern offices
Ring	Closed loop	Can break ring if one node fails	Medium	Token Ring networks
Full Mesh	Every-to-every	Very fault tolerant (alternate paths)	Very High	Military, ISP backbone
Tree	Hierarchical	Sections fail if branch node fails	Medium-High	Large organizations
Hybrid	Mixed	Depends on combination	High	Enterprise networks

EXAM FOCUS

- Full Mesh links formula: $n(n-1)/2$. For 5 nodes: $5 \times 4 / 2 = 10$ links.
- Star topology: central device failure = all down. One cable failure = only that node affected.
- Bus: single point of failure is the backbone cable. Terminators required at both ends.
- Ring uses TOKEN to control who can transmit — prevents collisions.
- Tree = hierarchical = combination of Star and Bus.

COMMON EXAM TRAPS

- Star vs Bus: In Star, if ONE cable breaks, only that node is affected. In Bus, if the backbone breaks, ALL nodes fail.
- Full Mesh is NOT used in LANs due to extreme cost — mainly used for critical backbone links.
- Hybrid is the MOST COMMON real-world topology — pure topologies exist mostly in theory/textbooks.

TOPIC 12

Network Devices

Hub • Switch • Router • Modem • Access Point • NIC • Gateway • Repeater

Overview — Why Different Devices?

Different devices operate at different OSI layers and perform different jobs. Understanding WHICH layer a device operates at tells you what it can and cannot do.

Device	OSI Layer	Uses	Key Job
Hub	Layer 1 – Physical	Electrical signal	Broadcasts data to ALL ports — dumb device
Switch	Layer 2 – Data Link	MAC Address	Forwards frames to SPECIFIC port — smart device
Router	Layer 3 – Network	IP Address	Routes packets BETWEEN different networks
NIC	Layer 1/2	MAC Address	Connects a device to a network (hardware card)
Modem	Layer 1	Analog/Digital signal	Converts digital data ↔ analog signal for telephone/cable lines
Access Point	Layer 2	MAC Address	Connects wireless devices to a wired network (WiFi)
Repeater	Layer 1	Electrical signal	Amplifies and retransmits signal to extend range
Gateway	Layer 4-7	Protocol translation	Connects networks using DIFFERENT protocols
Bridge	Layer 2	MAC Address	Connects two LAN segments, filters traffic by MAC

Hub — The Dumb Broadcaster

- A hub is a basic networking device that connects multiple computers in a LAN.
- When it receives data from one port, it **broadcasts (copies) it to ALL other ports** — whether or not the destination is on that port.
- This causes **collisions** — when two devices send data at the same time, signals collide and corrupt each other.
- Hub creates a single **collision domain** — all devices compete for the same bandwidth.
- Largely **obsolete** today — replaced by switches.

HUB BEHAVIOR:

[A] sends data to [C]

[A]--+

[B]---[HUB]--broadcasts to--> [A][B][C][D]

[C]--+

[D]--+

ALL devices receive the data even though only C needs it.

Wasteful and causes collisions.

Switch — The Smart Forwarder

- A switch learns which **MAC address** is connected to which port and builds a **MAC Address Table**.
- When it receives a frame, it forwards it **ONLY** to the port where the destination MAC address is connected.
- Each port is its own **collision domain** — no collisions between ports.
- Creates a single **broadcast domain** — broadcast messages still go to all ports.
- Switches are the **standard device** in modern LANs.

```
SWITCH BEHAVIOR (MAC Table built over time):
Switch's MAC Table:
+-----+-----+
| MAC | Port |
+-----+-----+
| MAC_A | 1 |
| MAC_B | 2 |
| MAC_C | 3 |
+-----+-----+
[A]--Port1--[SWITCH]--Port3--[C] (only C receives data)
[B]--Port2--|
A sends to C --> Switch checks table --> sends ONLY to Port 3.
```

Router — The Traffic Director

- A router connects **different networks** (e.g., your home LAN to the internet) and routes packets using **IP addresses**.
- Maintains a **Routing Table** — a map of known networks and the best paths to reach them.
- Performs **NAT** (Network Address Translation) — converts your private home IP to the public IP for internet communication.
- Each interface on a router is a separate **broadcast domain** AND collision domain.
- Your home WiFi router = Router + Switch + Access Point combined in one device.

```
ROUTER — CONNECTING DIFFERENT NETWORKS:
Network 1 (192.168.1.x) Network 2 (10.0.0.x)
[PC1: .1.5] [Server: .0.10]
[PC2: .1.6]---[ROUTER]---Internet---[Server2: .0.11]
[PC3: .1.7] |
| NAT: 192.168.1.x --> Public IP 49.x.x.x
Router reads DESTINATION IP ADDRESS to decide where to forward.
Switch reads MAC address (only within same network).
```

Modem — The Signal Converter

- **MOdulator-DEModulator** — converts digital data from your computer into analog signals for transmission over telephone/cable lines, and vice versa.
- **Modulation** = converting digital → analog (to send over phone line).
- **Demodulation** = converting analog → digital (when receiving from phone line).
- Needed because traditional telephone infrastructure transmits analog signals, but computers are digital.
- Modern fiber/cable internet uses modem variants for DOCSIS (cable) or ONT (fiber).

MODEM — SIGNAL CONVERSION:

Your PC ISP

[Digital: 10110101] --MODEM--> [Analog ~~~~] --> [ISP Network]
Modulate

[ISP Network] --> [Analog ~~~~] --MODEM--> [Digital: 10110101]
Demodulate

Without modem: digital computer CANNOT communicate over
traditional analog telephone lines.

NIC — Network Interface Card

- Hardware component (a card/chip) installed in a device that **physically connects it to a network**.
- Every NIC has a unique **MAC address** burned in at manufacture — this is the device's physical network identity.
- Can be wired (Ethernet port) or wireless (WiFi chip).
- Operates at both Layer 1 (physical transmission) and Layer 2 (MAC addressing).

Access Point (AP) — The Wireless Hub

- Connects **wireless devices** (phones, laptops) to a **wired network**.
- Acts as a bridge between the wireless WiFi network and the wired Ethernet network.
- Operates at Layer 2 — uses MAC addresses like a switch.
- Multiple APs can be placed in a building to extend WiFi coverage.
- Different from a router: AP only provides wireless ACCESS — it does NOT route between networks.

ACCESS POINT vs ROUTER:

[Laptop WiFi])

[Phone WiFi])---[ACCESS POINT]---[Ethernet]---[Switch]---[Router]

[Tablet WiFi])

AP only bridges wireless to wired.

Router handles internet routing.

Home WiFi router combines BOTH in one box.

EXAM FOCUS

- Hub = Layer 1, broadcasts to all. Switch = Layer 2, forwards by MAC to specific port.
- Router = Layer 3, routes by IP between DIFFERENT networks. Switch works within ONE network.
- Modem = MODulator/DEModulator. Converts digital ↔ analog.
- NIC contains the MAC address — it is the physical network identity of a device.
- Access Point provides wireless connectivity but does NOT route — that is the router's job.

COMMON EXAM TRAPS

- Router vs Switch: Switch uses MAC (same network). Router uses IP (different networks). Never confuse these.
- Hub vs Switch: Hub sends to ALL ports (causes collisions). Switch sends to SPECIFIC port (no collisions).
- Your 'WiFi router' at home is actually Router + Switch + Access Point combined — not a pure router.

TOPIC 13

Transmission Media

Guided (Wired) • Unguided (Wireless)

What is Transmission Media?

Transmission media is the physical path through which data travels from sender to receiver. It is broadly divided into **Guided (Wired)** — signal travels through a physical conductor — and **Unguided (Wireless)** — signal travels through air/space.

GUIDED (WIRED) MEDIA

• A. Twisted Pair Cable

- Two insulated copper wires twisted around each other to reduce **electromagnetic interference (EMI)** — the twisting causes interfering signals to cancel each other out.
- **UTP (Unshielded Twisted Pair)** — no extra shielding. Cheaper, most common. Used in Ethernet LANs (Cat5, Cat6 cables).
- **STP (Shielded Twisted Pair)** — has metal foil shielding around the pairs. Better interference protection, more expensive.
- Speed: up to 10 Gbps (Cat6a). Distance: up to ~100 metres before signal degrades.

TWISTED PAIR CABLE (cross-section view):

UTP: STP:

```
_____  
/ + - \ / [foil shield] \  
| wire wire | | [+wire -wire] |  
| twisted | | twisted+shield |  
\_____/ \_____/
```

Twisting cancels out EMI from nearby cables/devices.

Cat5e: up to 1 Gbps | Cat6: up to 10 Gbps (short dist)

• B. Coaxial Cable

- Central copper conductor surrounded by insulation, metallic mesh shield, and outer plastic jacket.
- Much better noise resistance than twisted pair due to the shield.
- Used for: cable TV (CATV), older Ethernet networks (Thinnet/Thicknet), internet via cable provider.
- Speed: up to ~10 Gbps. Distance: up to several kilometres.

COAXIAL CABLE (cross-section):

[Outer jacket (plastic)]

[Metallic mesh shield]

[Insulation (dielectric)]

[Central copper conductor]

4 layers from outside to inside:

Outer Jacket --> Shield --> Insulation --> Copper Core

Shield protects central wire from EMI.

• C. Fiber Optic Cable

- Transmits data as **pulses of light** through thin glass or plastic fibers, instead of electrical signals.
- No electromagnetic interference (light is not affected by EMI).
- **Single-mode fiber** — thin core, light travels straight. Long distances (up to 100 km). Used for long-haul internet backbone.
- **Multi-mode fiber** — wider core, light bounces at angles (multiple paths). Shorter distances (up to 2 km). Used within buildings/campuses.
- Fastest transmission medium. Speed: up to **terabits per second**. Not affected by electrical interference.
- Disadvantage: Expensive, fragile, harder to install, difficult to splice.

FIBER OPTIC — LIGHT TRANSMISSION:

Single-mode: Multi-mode:

=====> ==~===>

(straight path) (bouncing paths)

Long distance Short distance

8-10 micron core 50-62.5 micron core

Structure of fiber cable:

[Outer jacket] --> [Buffer coating] --> [Cladding] --> [Core (glass)]

Light reflects inside core due to total internal reflection.

UNGUIDED (WIRELESS) MEDIA

• A. Radio Waves

- Electromagnetic waves in frequency range **3 KHz to 1 GHz**.
- Omnidirectional — travel in all directions, can penetrate walls.
- Used for: WiFi, AM/FM radio, Bluetooth, cellular networks (4G/5G).
- Disadvantage: Susceptible to interference; security concerns (signal goes everywhere).

• B. Microwave

- Frequency range **1 GHz to 300 GHz**. High frequency = high data rate.
- **Terrestrial microwave** — line-of-sight transmission between towers on Earth. Towers placed every 25-30 km.
- **Satellite microwave** — signal sent to satellite in orbit; satellite relays it back to Earth. Very long distances.
- Directional — antennas must be precisely aligned. Cannot pass through buildings.

• C. Infrared

- Very short range communication (up to a few metres). Cannot penetrate walls.
- Used for: TV remote controls, short-range device communication (old IrDA standard).
- Disadvantage: Line-of-sight required, very short range, blocked by obstacles.

Media Type	Signal Type	Speed	Distance	Interference	Example Use
Twisted Pair (UTP)	Electrical	Up to 10 Gbps	~100 m	Medium (EMI)	Office LAN, home Ethernet
Coaxial Cable	Electrical	Up to 10 Gbps	Several km	Low (shielded)	Cable TV, older Ethernet
Fiber Optic	Light pulses	Tbps range	100+ km	None (immune to EMI)	Internet backbone, ISP links
Radio Waves	EM waves	Varies	km range	High	WiFi, Bluetooth, 4G/5G

Microwave	EM waves	High	25-30 km/tower	Medium	Satellite TV, backbone links
Infrared	EM waves	Low	< 5 m	High (blocked)	TV remote, IrDA

EXAM FOCUS

- Fiber optic = FASTEST, immune to EMI, most expensive, uses light.
- Twisted pair = MOST COMMON for LAN. Twisting reduces EMI/crosstalk.
- UTP vs STP: UTP has no shield (cheaper). STP has foil shield (better noise protection).
- Single-mode fiber = long distance (100km), thin core. Multi-mode = short distance (2km), wide core.
- Guided = wired (signal has a physical conductor). Unguided = wireless (signal through air).
- Infrared: line-of-sight only, cannot pass walls, very short range.

COMMON EXAM TRAPS

- Fiber optic is immune to EMI because it uses LIGHT not electricity — electromagnetic fields have no effect on light.
- Microwave requires LINE-OF-SIGHT — buildings/mountains block the signal. Radio waves can go around/through obstacles.

Switching Techniques

Circuit Switching • Packet Switching • Message Switching

What is Switching?

Switching is the technique used to establish a communication path between sender and receiver and to forward data through a network. Different methods handle this differently — with major differences in efficiency, speed, and resource usage.

1. Circuit Switching

- A **dedicated physical path** is established between sender and receiver **BEFORE** data transmission begins, and this path is held for the **entire duration of the communication**.
- Three phases: **Circuit Establishment** → **Data Transfer** → **Circuit Teardown**.
- Resources (bandwidth) are reserved along the entire path — even when no data is being sent (idle time wastes reserved bandwidth).
- **Advantage:** Guaranteed bandwidth, constant delay, no queuing — excellent for real-time voice.
- **Disadvantage:** Wasteful (idle periods still occupy reserved circuit), slow setup time, inflexible.
- **Classic example:** Traditional telephone (PSTN) calls — a dedicated line is held for the whole call.

CIRCUIT SWITCHING:

Step 1 – ESTABLISH dedicated path:

[A]---[Router1]---[Router2]---[Router3]---[B]
(reserved) (reserved) (reserved)

Step 2 – DATA TRANSFER along fixed path:

[A] =====> [B]
(entire path dedicated to A-B only)

Step 3 – TEARDOWN path after call ends.

Even if A is silent for 10 seconds, the path is **STILL** reserved and cannot be used by others. --> Wasteful!

2. Packet Switching

- Data is broken into small chunks called **packets**. Each packet travels **independently** through the network, possibly taking different routes.
- No dedicated path is reserved — routers decide the **best path for each packet dynamically**.
- Each packet contains: **Header** (destination IP, sequence number) + **Payload** (actual data).
- **Two types:**
 - Datagram Packet Switching** — each packet routed independently, may arrive out of order.
 - Virtual Circuit Packet Switching** — a logical path (not physically dedicated) is set up; all packets follow same route but resources not fully reserved.
- **Advantage:** Efficient (links shared among many users), robust (packets reroute around failures), scalable.
- **Disadvantage:** Variable delay (packets may queue at routers), packets can arrive out of order (must be reassembled).
- **Classic example:** The Internet — all web browsing, email, video streaming uses packet switching.

PACKET SWITCHING:

Message: 'HELLO' split into packets:

Packet 1: [Header: dest=B, seq=1 | 'HE']

Packet 2: [Header: dest=B, seq=2 | 'LL']

Packet 3: [Header: dest=B, seq=3 | 'O']

[A] sends all 3 packets independently:

Packet1: [A]--R1--R2-----[B] (route via R2)

Packet2: [A]--R1--R3--R4---[B] (route via R3,R4)

Packet3: [A]--R1--R2-----[B] (route via R2)

Packets may ARRIVE OUT OF ORDER at B.

B reassembles in correct order using sequence numbers.

Links R1-R2 also carry OTHER users' packets simultaneously.

3. Message Switching

- The entire message (not broken into packets) is sent from node to node using a **store-and-forward** approach.
- **Store-and-forward** — each intermediate node receives the COMPLETE message, stores it, then forwards it to the next node when the outgoing link is free.
- **Advantage:** No dedicated path needed. Can handle traffic efficiently when links are shared.
- **Disadvantage:** Large messages occupy router storage for long periods. High delay — must wait for entire message to be received before forwarding. Not suitable for real-time communication.
- Historically used in **telegraph networks** and early email systems. Largely replaced by packet switching.

MESSAGE SWITCHING (store-and-forward):

Full message = 'HELLO WORLD HOW ARE YOU'

[A] sends entire message to [R1]

[R1] STORES full message --> waits for R2 link --> FORWARDS

[R2] STORES full message --> waits for R3 link --> FORWARDS

[R3] STORES full message --> waits for B link --> FORWARDS

[B] receives complete message

If message is 10 GB --> routers must store 10 GB each!

vs Packet Switching: only small packets (e.g., 1 KB) stored.

Feature	Circuit Switching	Packet Switching	Message Switching
Path	Dedicated physical path	No dedicated path (dynamic)	No dedicated path (store-forward)
Resource reservation	Yes — reserved throughout	No — shared as needed	No
Data unit	Continuous bit stream	Packets (small chunks)	Entire message
Delay	Low, constant	Variable (queuing)	High (store-and-forward)
Efficiency	Low (idle bandwidth wasted)	High (shared links)	Medium
Order of delivery	In order (fixed path)	May arrive out of order	In order
Example	Phone call (PSTN)	Internet (TCP/IP)	Telegraph, early email

EXAM FOCUS

- Circuit Switching: dedicated path, 3 phases (establish-transfer-teardown), wastes bandwidth when idle.
- Packet Switching: no dedicated path, data broken into packets, each routed independently, internet uses this.
- Message Switching: store-and-forward, entire message stored at each node, not suitable for real-time.
- Packet Switching is MOST EFFICIENT for internet-type traffic. Circuit Switching best for real-time voice.

COMMON EXAM TRAPS

- Circuit vs Packet: Circuit has dedicated reserved bandwidth (wasteful). Packet shares bandwidth dynamically (efficient).
- Packet Switching can deliver packets OUT OF ORDER — receiver reassembles using sequence numbers.
- Message Switching vs Packet Switching: Message = whole message forwarded. Packet = broken into small pieces forwarded.

Network Performance Metrics

Bandwidth • Latency • Throughput • Jitter • Packet Loss

Why Measure Network Performance?

A network's quality depends not just on whether data arrives, but on HOW FAST and HOW RELIABLY it arrives. These metrics quantify network performance and help diagnose problems.

1. Bandwidth

- **Definition:** The **maximum amount of data** that can be transmitted over a network link in a given time period.
- **Measured in:** bits per second (bps), Kbps, Mbps, Gbps, Tbps.
- **Analogy:** bandwidth is like the **width of a road** — a 4-lane highway (high bandwidth) can carry more cars (data) at once than a 1-lane road (low bandwidth).
- Bandwidth is a **capacity/limit** — the maximum possible, not what you always get.
- ISPs advertise 'up to 100 Mbps' — this is the bandwidth ceiling; actual speed may be lower.

2. Latency (Delay)

- **Definition:** The **time taken for a packet to travel** from source to destination.
- **Measured in:** milliseconds (ms). Lower is better.
- **Analogy:** latency is like the **travel time on the road** — even a wide road has travel time. A road to the next city takes longer than a road to the next street.
- **Components of latency:**
 - Propagation delay** — time for signal to physically travel the distance (speed of light/electricity).
 - Transmission delay** — time to push all bits onto the wire (depends on bandwidth).
 - Processing delay** — time routers take to examine and forward a packet.
 - Queuing delay** — time packet waits in router queue if link is busy.
- **RTT (Round Trip Time)** — time for a packet to go to destination AND come back. What 'ping' measures.

3. Throughput

- **Definition:** The **actual amount of data successfully transferred** per unit time in practice.
- **Measured in:** bps, Mbps, etc. Always $\leq$ Bandwidth.
- **Analogy:** if bandwidth is the road width (capacity), throughput is the **actual number of cars that passed** in an hour (accounting for traffic jams, accidents, and slow drivers).
- **Why throughput < bandwidth:** packet loss forces retransmissions, protocol overhead (headers), network congestion, interference.

4. Jitter

- **Definition:** **Variation in latency** — when packets arrive at irregular intervals instead of a steady pace.
- **Example:** packet 1 arrives in 10ms, packet 2 in 50ms, packet 3 in 15ms — jitter = inconsistency.
- **Critical for real-time applications:** high jitter causes choppy audio/video in video calls.
- **Jitter buffers** — used in VoIP/video apps to absorb jitter by slightly delaying playback.

5. Packet Loss

- **Definition:** Percentage of packets that are sent but never arrive at the destination.

- Causes: network congestion (routers drop packets when queues overflow), corruption (interference), hardware failure.
- TCP handles packet loss by retransmitting lost packets. UDP does not — lost UDP packets are gone.

BANDWIDTH vs THROUGHPUT vs LATENCY – ROAD ANALOGY:

BANDWIDTH: ===4-lane highway=== (capacity: 4 cars side by side)

THROUGHPUT: ==2 cars actually== (only 2 passing due to traffic)

LATENCY: |<---travel time--->| (how long to get from A to B)

JITTER: car1 arrives 10min, car2 arrives 50min, car3 10min
(inconsistent arrival = jitter)

SUMMARY:

Bandwidth = max capacity of link (bps)

Throughput = actual data delivered (always $\leq$ bandwidth)

Latency = delay from source to destination (ms)

Jitter = variation in latency (ms)

Packet loss= % of packets dropped

Metric	Unit	What it Measures	Good Value	Bad Impact if Poor
Bandwidth	Mbps/Gbps	Max data capacity of link	Higher better	Slow downloads/uploads
Latency	Milliseconds (ms)	Delay from source to destination	Lower better	Lag in gaming/calls
Throughput	Mbps	Actual data delivered	Higher better ($\leq$ bandwidth)	Slow actual transfer
Jitter	ms (variation)	Consistency of delay	Near 0 ms	Choppy video/voice
Packet Loss	%	Packets dropped	Near 0%	Retransmission, bad call quality

EXAM FOCUS

- Bandwidth = maximum capacity. Throughput = actual delivered rate. Throughput is ALWAYS $\leq$ Bandwidth.
- Latency = delay. RTT = Round Trip Time = latency x2 (there and back). Ping measures RTT.
- Jitter = variation in latency. Causes choppy audio/video in real-time communication.
- High bandwidth does NOT guarantee low latency — they measure different things.

COMMON EXAM TRAPS

- Bandwidth vs Throughput: Bandwidth is theoretical max (road capacity). Throughput is what you actually get (cars that passed). Never say they are the same.
- High bandwidth with high latency still gives bad gaming experience — latency matters more than bandwidth for real-time apps.

TOPIC 16

Internet Basics

WWW • ISP • URL • Web Browser • DNS • HTTP • IP (surface)

What is the Internet?

The **Internet** is a global system of interconnected computer networks that communicate using the standardized **TCP/IP protocol suite**. It is a 'network of networks' — millions of smaller networks (home, office, ISP, data center) all connected together.

Internet vs World Wide Web (WWW)

	Internet	World Wide Web (WWW)
What it is	Global infrastructure of connected networks	A SERVICE that runs ON TOP of the internet — a collection of web pages and sites
Analogy	The roads and highways of a country	The shops, houses, and destinations built along those roads
Created by	ARPANET (1960s-1970s), evolved over decades	Tim Berners-Lee in 1989
Includes	Email, FTP, streaming, gaming, WWW, VoIP, etc.	Only web pages accessed via HTTP/HTTPS in browsers
Example	All internet traffic	Browsing google.com, wikipedia.org

Key fact: The WWW is a SUBSET of the internet. The internet is much larger — it includes email, FTP, online gaming, VoIP, etc., not just web browsing.

ISP — Internet Service Provider

- An **ISP** is a company that provides internet access to homes and businesses. Examples: Jio, Airtel, BSNL, ACT Fibernet.
- Your device connects to the ISP's network; ISP connects you to the global internet.
- **Tiers of ISPs:**
 - Tier 1 ISPs** — Largest global ISPs that own the internet backbone (AT&T, NTT). They peer with each other for free.
 - Tier 2 ISPs** — Regional ISPs that buy access from Tier 1 and sell to Tier 3.
 - Tier 3 ISPs** — Local ISPs (Jio, Airtel) that buy from Tier 2 and sell directly to end users.

ISP HIERARCHY:

```
[Tier 1: Global backbone] <--peering--> [Tier 1: Global backbone]
| |
[Tier 2: Regional ISP] [Tier 2: Regional ISP]
| |
[Tier 3: Jio] [Tier 3: Airtel]
| |
[Your home] [Your friend's home]
```

URL — Uniform Resource Locator

A **URL** is the complete address used to locate a specific resource on the internet.

URL STRUCTURE:

`https://www.example.com:443/page/article?id=5#section2`

|_____| |_____| | | |_____| |_____| |_____|

Scheme Domain name Port Path/Resource Query Fragment

Scheme : protocol used (http, https, ftp)

Domain : human-readable name of the server (example.com)

Port : service port number (443 for HTTPS; often omitted if default)

Path : location of specific resource on server

Query : additional parameters (?key=value)

Fragment: specific section within a page (#section2)

DNS — Domain Name System (Expanded)

- DNS converts **domain names** (google.com) to **IP addresses** (142.250.x.x).
- **DNS Hierarchy:**
 - Root DNS servers** — top of hierarchy; know addresses of TLD servers. 13 root server clusters worldwide.
 - TLD (Top Level Domain) servers** — handle .com, .org, .in, .edu etc.
 - Authoritative DNS servers** — hold actual IP for specific domains.
 - Recursive Resolver** — your ISP's DNS server; queries on your behalf.

DNS RESOLUTION PROCESS (first time, nothing cached):

Your PC --> [Recursive Resolver / ISP DNS]

|

v

'Where is google.com?'

[Root Server]: 'I don't know, ask .com TLD server'

|

[.com TLD Server]: 'Ask Google's authoritative server'

|

[Google Auth DNS]: 'google.com = 142.250.193.46'

|

[Recursive Resolver caches result and returns to your PC]

|

Your PC uses 142.250.193.46 to connect to Google.

HTTP and HTTPS (Expanded)

- **HTTP (HyperText Transfer Protocol)** — rules for how web browsers request and receive web pages.
- **HTTP Methods:**
 - GET** — request data from server (loading a webpage).
 - POST** — send data TO server (submitting a form, login).
 - PUT** — update existing data on server.
 - DELETE** — delete data on server.
- **HTTP Status Codes:**
 - 200 OK** — request succeeded.

301 Moved Permanently — resource has new URL.

404 Not Found — resource does not exist.

500 Internal Server Error — server-side problem.

- **HTTPS** = HTTP + TLS/SSL encryption. Padlock icon in browser confirms HTTPS. Protects login credentials, payment info.

Web Browser — How It Works

WHAT HAPPENS WHEN YOU TYPE 'www.google.com' AND PRESS ENTER:

1. Browser checks its own DNS cache --> not found
2. Asks OS DNS cache --> not found
3. Asks ISP Recursive Resolver --> queries DNS hierarchy
4. Gets IP: 142.250.193.46
5. ARP: find router's MAC (to send packet out of local network)
6. TCP 3-Way Handshake with Google's server:
Your PC --SYN--> Google
Your PC <--SYN-ACK-- Google
Your PC --ACK--> Google
7. TLS Handshake (for HTTPS) -- establishes encryption
8. HTTP GET request sent to Google
9. Google sends HTTP 200 OK + HTML page
10. Browser renders HTML, requests images/CSS/JS (more HTTP GETs)
11. Page displayed

EXAM FOCUS

- Internet = global network infrastructure. WWW = web pages only, a subset of the internet.
- URL structure: Scheme://Domain:Port/Path?Query#Fragment.
- HTTP methods: GET (retrieve), POST (send), PUT (update), DELETE (remove).
- HTTP 200=OK, 404=Not Found, 500=Server Error. Memorize these three.
- DNS hierarchy: Root → TLD (.com/.in) → Authoritative → Recursive Resolver.
- HTTPS = HTTP + TLS/SSL. Port 80 = HTTP. Port 443 = HTTPS.

COMMON EXAM TRAPS

- Internet ≠ WWW. Internet includes email, gaming, FTP, VoIP, etc. WWW is only web pages.
- DNS uses both UDP (port 53) for regular queries and TCP (port 53) for large transfers/zone transfers.
- Tim Berners-Lee invented the WWW (1989), NOT the internet itself.

TOPIC 17

Network Security Basics

Threats • Firewall • Encryption • VPN • Protection Techniques

Why Network Security?

As networks connect billions of devices globally, they also create pathways for attackers to steal data, disrupt services, or take control of systems. Network security is the set of policies, tools, and practices that protect network resources from unauthorized access and attacks.

Common Threats and Attacks

Threat	What It Does	Example
Virus	Malicious code that attaches itself to files and self-replicates when the file is run. Spreads by copying itself to other files.	ILOVEYOU virus (spread via email attachments)
Worm	Malicious program that self-replicates and spreads ACROSS NETWORKS without needing to attach to a file — travels on its own.	WannaCry ransomware (spread via network vulnerability)
Trojan Horse	Malicious code disguised as legitimate software. Does not self-replicate but allows attackers to access the system once installed.	Fake antivirus software that steals data
Ransomware	Encrypts victim's files and demands payment (ransom) to restore access.	WannaCry, REvil
Phishing	Fake emails/websites that trick users into revealing passwords, credit card numbers, or personal info.	Fake bank email asking to 'verify your account'
Spyware	Software that secretly collects user information (keystrokes, browsing habits) and sends it to attacker.	Keyloggers, adware
DoS / DDoS	Denial of Service: floods a server with so many requests it cannot serve real users. DDoS = Distributed DoS (many machines).	Flooding a web server with millions of fake requests
Man-in-the-Middle	Attacker secretly intercepts and possibly alters communication between two parties without either knowing.	Intercepting unencrypted WiFi traffic
SQL Injection	Attacker inserts malicious SQL code into input fields to manipulate a database.	Login form accepting ' OR 1=1 -- to bypass password
Spoofing	Impersonating another device/user by faking their IP or MAC address.	IP spoofing to bypass firewall rules

Firewall — The Network Gatekeeper

- A **firewall** is a security device (hardware or software) that **monitors and controls incoming and outgoing network traffic** based on predefined security rules.
- Acts as a barrier between a trusted internal network and untrusted external networks (internet).

- **Packet Filtering Firewall** — inspects each packet's header (source/destination IP, port, protocol) and allows or blocks based on rules. Simple and fast but cannot inspect packet content.
- **Stateful Inspection Firewall** — tracks the STATE of connections. Knows which packets belong to established connections vs suspicious new ones.
- **Application Layer Firewall (Proxy Firewall)** — operates at Layer 7. Can inspect actual content (HTTP, FTP). Most secure but slowest.
- **Next-Generation Firewall (NGFW)** — combines all above + deep packet inspection, intrusion prevention, VPN support.

```

FIREWALL POSITION IN NETWORK:
INTERNET (untrusted)
|
[FIREWALL] <-- inspects ALL traffic, applies rules
|
+-----+-----+
| INTERNAL NETWORK (trusted) |
| [PC1] [PC2] [Server] [PC3] |
+-----+-----+
Firewall RULES example:
ALLOW: incoming TCP port 80 (web) from anywhere
ALLOW: incoming TCP port 443 (HTTPS) from anywhere
BLOCK: incoming TCP port 23 (Telnet) from anywhere
BLOCK: outgoing traffic to known malicious IPs

```

Encryption

- **Encryption** = converting readable data (plaintext) into unreadable form (ciphertext) using a mathematical algorithm and a **key**. Only someone with the correct key can decrypt it back.
- **Symmetric Encryption** — SAME key used for encryption and decryption. Fast but key must be shared securely. Example: AES.
- **Asymmetric Encryption (Public Key)** — TWO keys: **Public key** (anyone can use to encrypt) and **Private key** (only owner uses to decrypt). Used in HTTPS/TLS. Example: RSA.
- **SSL/TLS** — protocols used to encrypt HTTPS connections between browser and server.

```

SYMMETRIC vs ASYMMETRIC ENCRYPTION:
SYMMETRIC (same key):
[Plaintext] --[Key A]--> [Ciphertext] --[Key A]--> [Plaintext]
Encrypt with Key A Decrypt with SAME Key A
Problem: How do you securely share Key A?
ASYMMETRIC (key pair):
[Plaintext] --[Public Key]--> [Ciphertext] --[Private Key]--> [Plaintext]
Anyone encrypts with Only owner decrypts with
your PUBLIC key their PRIVATE key
Public key can be shared freely -- no security risk.

```

VPN — Virtual Private Network

- A **VPN** creates an **encrypted tunnel** between your device and a VPN server, routing all your internet traffic through it.

- **Benefits:** Hides your real IP address. Encrypts traffic even on public WiFi. Allows access to geo-restricted content. Lets remote workers securely access company intranet.
- **How it works:** Your device encrypts data → sends encrypted tunnel to VPN server → VPN server decrypts and forwards to internet. Websites see VPN server's IP, not yours.

```
VPN TUNNEL:
WITHOUT VPN:
[Your PC] ---clear text---> [Internet] --> [Website]
(ISP can see everything, website sees your real IP)
WITH VPN:
[Your PC] ==encrypted tunnel==> [VPN Server] --> [Website]
(ISP sees encrypted data only, website sees VPN server IP)
```

Other Security Concepts

- **IDS (Intrusion Detection System)** — monitors network traffic for suspicious activity and ALERTS administrators. Passive — detects but does not block.
- **IPS (Intrusion Prevention System)** — like IDS but ACTIVELY blocks suspicious traffic. Inline device.
- **DMZ (Demilitarized Zone)** — a network segment between internal and external network where public-facing servers (web servers) are placed — separate from internal trusted network.
- **MAC Filtering** — router/AP allows only specific MAC addresses to connect to the network.
- **Two-Factor Authentication (2FA)** — requires two forms of verification (password + OTP) before granting access.
- **Antivirus/Antimalware** — software that scans files and memory for known malicious patterns and removes them.

```
NETWORK SECURITY LAYERS (Defence in Depth):
[Internet]
|
[Firewall] -- blocks unauthorized traffic
|
[IPS] -- detects and blocks intrusions
|
[DMZ] -- [Web Server] [Email Server] (semi-public zone)
|
[Internal Firewall]
|
[Internal Network]
[PC1] [PC2] [Database Server]
(most protected zone -- never directly accessible from internet)
```

EXAM FOCUS

- Virus = attaches to files, needs host. Worm = spreads on its own across network, no host file needed.
- Trojan = disguised as legitimate software. Ransomware = encrypts data, demands ransom.
- Phishing = social engineering via fake emails/websites to steal credentials.
- Firewall = monitors and filters traffic based on rules. Acts as gatekeeper between trusted and untrusted networks.
- Symmetric encryption = same key both ways (fast, AES). Asymmetric = public key encrypts, private key decrypts (RSA, used in HTTPS).
- DoS = single attacker floods server. DDoS = Distributed DoS from many machines simultaneously.

COMMON EXAM TRAPS

- Virus vs Worm: Virus needs to ATTACH to a file and requires user action to spread. Worm travels INDEPENDENTLY across the network.
- IDS vs IPS: IDS only DETECTS and alerts (passive). IPS PREVENTS/blocks (active inline device).
- Firewall vs Antivirus: Firewall filters NETWORK traffic at border. Antivirus scans files/memory ON the device.

MASTER QUICK REFERENCE

Topics 10-17 — All Critical Facts

Network Types at a Glance

Type	Range	Example
PAN	~10 m	Bluetooth earbuds
LAN	1 building	Office network
MAN	1 city	City cable network
WAN	Global	Internet
Intranet	Internal org only	Company HR portal
Extranet	Org + trusted partners	Vendor portal
Internet	Global public	Everything

Topology Comparison

Topology	Key Feature	Single Point of Failure	Cost
Bus	Single backbone	Backbone cable	Low
Star	Central hub/switch	Central hub/switch	Medium
Ring	Closed loop, token	One node can break ring	Medium
Full Mesh	Every-to-every, $n(n-1)/2$ links	Highly fault tolerant	Very High
Tree	Hierarchical, Star+Bus	Branch node	Medium-High
Hybrid	Mixed topologies	Depends	High

Network Devices Summary

Device	OSI Layer	Identifies By	Primary Job
Hub	L1	Nothing (broadcasts all)	Connect devices, broadcast signal
Switch	L2	MAC Address	Forward frames to correct port only
Router	L3	IP Address	Route packets between different networks
Modem	L1	Analog/Digital	Convert digital<->analog for ISP connection
Access Point	L2	MAC Address	Connect wireless devices to wired network

NIC	L1/L2	MAC Address (built-in)	Physical network connection for a device
Repeater	L1	Nothing	Amplify and extend signal range
Gateway	L4-7	Protocol	Connect networks with different protocols

Switching Techniques Summary

Technique	Path	Data Unit	Delay	Example
Circuit Switching	Dedicated reserved path	Continuous stream	Low, constant	Telephone (PSTN)
Packet Switching	Dynamic, per packet	Packets (small chunks)	Variable	Internet
Message Switching	Store-and-forward	Whole message	High	Telegraph, early email

Transmission Media Summary

Media	Type	Speed	Distance	Best For
Twisted Pair (UTP/STP)	Wired	Up to 10 Gbps	~100 m	LAN, home Ethernet
Coaxial	Wired	Up to 10 Gbps	Several km	Cable TV, old Ethernet
Fiber Optic	Wired	Tbps+	100+ km	Internet backbone, ISP
Radio Waves	Wireless	Varies	km	WiFi, Bluetooth, 4G/5G
Microwave	Wireless	High	25-30 km/tower	Satellite, long-distance
Infrared	Wireless	Low	< 5 m	TV remote, IrDA

Performance Metrics at a Glance

Metric	Measures	Unit	Rule
Bandwidth	Maximum link capacity	Mbps/Gbps	Higher = better capacity
Throughput	Actual data delivered	Mbps	Always ≤ bandwidth
Latency	Delay source to destination	ms	Lower = less delay
Jitter	Variation in latency	ms	Near 0 = smooth stream
Packet Loss	Packets dropped	Percentage	Near 0% = reliable

Security Threats Quick Reference

Threat	Key Distinguisher
--------	-------------------

Virus	Attaches to file, needs user action to spread
Worm	Spreads independently across network, no host file
Trojan	Disguised as legit software, no self-replication
Ransomware	Encrypts data, demands payment to decrypt
Phishing	Fake email/site tricks user into revealing credentials
DoS/DDoS	Floods server with requests; DDoS = many attacking machines
Man-in-Middle	Silently intercepts communication between two parties
Spyware	Secretly collects user data without knowledge